

Mechanism Black Box Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	mechanism-black-box

Big idea: Scientific explanation often goes beyond prediction by proposing mechanisms.

One-Minute Teacher Brief

Students infer the mechanism inside a sealed system by testing inputs and outputs, then compare competing explanations.

Student Prompt

Inspect Mechanism Black Box Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Mechanism Black Box Stimulus A	Student-facing stimulus 1: Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Mechanism Black Box Stimulus B	Student-facing stimulus 2: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Mechanism Black Box Reveal / Twist	Student-facing stimulus 3: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific explanation often goes beyond prediction by proposing mechanisms.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Mechanism Black Box Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Mechanism Black Box Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	What is added by knowing a mechanism?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how scientific explanation often goes beyond prediction by proposing mechanisms.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: What is added by knowing a mechanism?

Setup Checklist

- Use a hidden input-output rule that several mechanisms could initially explain.
- Prepare at least one test input that separates competing explanations.

Example Stimuli or Materials

- Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples.
- Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card.

Resources To Use

- Resource pack: <resources/mechanism-black-box/index.html>
- Card deck CSV: <resources/mechanism-black-box/cards.csv>
- Visual prompt: <resources/mechanism-black-box/visual-prompt.svg>
- Input-output testing sheet.
- Mechanism diagram card with spaces for rule, prediction, and disconfirming test.

Run Sequence

1. Introduce a black box that produces outputs from inputs according to a hidden rule.
2. Groups test inputs and record outputs.
3. Students propose a mechanism that could explain the pattern.
4. Groups test a new input to discriminate between mechanisms.
5. Debrief why explanation requires more than curve-fitting.

Debrief Moves

- Knowledge question: What is added by knowing a mechanism?
- Can a model predict well without explaining?

- How do scientists choose between competing unseen explanations?

Teacher Quick Script

- A pattern that predicts is useful, but a mechanism says why.
- What test would make one explanation fail?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...